Smaran Teja

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Experience

Teaching Assistant: Logic and Computation, Northeastern University *Jan 2026 – Present*

- Co-developed **OCaml** library for developing test suites to run submission code against and provide immediate results
- Created **Java** HW assignment for developing property-based specifications using jqwik
- Held midterm review session with **30+** students, performing code reviews of sample student implementations

Research co-op: Machine Learning Code Profiler, Northeastern *PRL* *Sep 2025 – Present*

- Designed a novel multi-threaded profiling algorithm to analyze *Roulette* Probabilistic Programming Language (PPL)
- Developed robust **Racket** algorithm to generate profiling results and HTML visualization tool in **Python**
- Optimized various performance and memory metrics to increase efficiency of profiler
- Evaluated multiple new heuristics for identifying possible areas of performance issues in PPLs
- Presented findings and ongoing research report at the NEU Probabilistic Programming Laboratory

Course Development Instructor: CS2100, Khoury College of Computer Sciences *May – Sep 2025*

- Led the development of course materials and infrastructure for new intro CS course
- Built **Python** autograders via a *GitHub Action* with support for mutation testing and HTML report generation
- Developed **TypeScript** code to display results/feedback on *Pawtograder* website.
- Implemented dependency caching algorithm across runs that reduced server usage by **75%**, and provides results in <10s
- Onboarded new professors to the grading interface and autograder infrastructure

Teaching Assistant: Fundamentals 1, Northeastern University *Sep – Dec 2024*

- **Lead TA** for a lab section with 30+ students, administered in-person code walkthroughs and peer evaluation weekly
- Held weekly office hours to perform extensive grading reviews for HWs and exams and reinforce lecture concepts

Projects

P2P distributed file sharing system [code] *Mar 2026 – Present*

- Developed a distributed file sharing network in **Java** using Chord protocol and TCP socket communication
- Engineered fault tolerant architecture to create a robust Distributed Hash Table

***FeedBot:* automated feedback tool [code]** *June – Dec 2024*

- Lead developer and maintainer for ***FeedBot*** tool, providing natural language feedback based on code design recipe.
- Now used campus-wide for intro CS HWs, reaching **500+ students** per year
- Deployed user-friendly website using **Python** and **Jinja** to parse and display problem-wise FeedBot results to students
- Co-authored a research report: "*FeedBot: Formative Design Feedback on Programming Assignments*"

***Synth*: LLM-backed program synthesis Python library and research project [code]** *Jan – June 2024*

- Developed an LLM-based Program Synthesis tool using **Python** and **OpenAI APIs**
- Developed efficient caching and re-querying mechanism in case of failed synthesis to reduce API usage by **80%**
- Performed user testing for tool with **100+ live testers** in lab sections for a CS course
- Authored a (draft) research paper: "*Programming with non-algorithmic specifications*"

Education

Northeastern University: Khoury College of Computer Sciences, Boston, MA *Sep 2023 – Present*

Candidate for Bachelor of Science in Computer Science, Expected May 2027

GPA: 3.8/4.0 (Dean's list, multiple semesters)

Relevant Courses: **Software engineering** | **Distributed systems** | **Programming languages** | ***Accelerated* Fundamentals of Computer Science 1 & 2** | Algorithms and Data | Object Oriented Design | Computer Systems | *Intensive* Math Reasoning

Technical Knowledge

Languages: Java | Python | C/C++ | JavaScript/TypeScript | Racket | OCaml | HTML/CSS

Applications/software: Git | VS Code | IntelliJ IDEA | MongoDB | Android Studio | Eclipse | Firebase | NetBeans | MySQL | React | Node.js | Jinja | Flask | DrRacket

Interests: Formula 1, Programming languages, Cricket, Dogs